

Homework 1

Powers, factorising and algebraic fractions · about 60 minutes

Each question graded N0–E8

Show all working. Leave answers with positive indices and fractions in simplest form. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

1 Write $\frac{1}{x^3}$ using a negative index. [A]

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2 Ana says $(a^2)^3 = a^5$. Is she right? Give a reason. [A]

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3 Kai says $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$. Show why he is wrong. [A]

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4 Which is bigger: 2^0 or 0^2 ? [A]

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QUESTION ONE

(a) Simplify each expression, leaving your answer with positive indices.

(i) $\frac{(4a)^2}{2a^{-3}}$ [A]

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(ii) $\sqrt{81m^{20}n^6}$ [A]

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(b) Factorise completely $10x^3y^2 - 25x^2y^3$. [A]

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(c) Simplify fully, leaving your answer with positive indices. [M]

$\left(\frac{27x^6}{x^{12}}\right)^{-\frac{1}{3}}$

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- (d) Sam was asked to factorise completely (i) $2x^3 - 50x$ and (ii) $3x^2 + 5x - 12$. **Sam's answer has 2 errors.**
Find each one, fix it, and say why it loses marks. **[M]**

Sam's answer

(i) $2x^3 - 50x = 2x(x^2 - 25) = 2x(x - 25)(x + 25)$

(ii) $3x^2 + 9x - 4x - 12 = 3x(x + 3) - 4(x - 3)$ so it won't factorise

- (e) Box A has sides $(x + 3)$, $(x - 2)$ and $(x + 1)$ cm. Box B has sides x , $(x + 1)$ and $(x + 2)$ cm, where $x > 2$.

- (i) Show that Box B always has the larger volume. **[E]**

- (ii) Find the value of x for which Box B's volume is 36 cm^3 more than Box A's. **[E]**

QUESTION TWO

- (a) Simplify $\frac{x^2 + 4x - 21}{2x + 14}$. **[A]**

- (b) Simplify $\frac{3x^2 + 7x + 2}{9x^2 - 1}$. **[A]**

- (c) Write $\frac{2x}{x + 4} - \frac{x - 3}{x - 1}$ as a single fraction in its simplest form. **[M]**

(d) Simplify $\frac{x^2 - 9}{4x} \div \frac{x^2 + x - 6}{8x^2}$. [M]

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(e)

(i) Show that $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$. [M]

(ii) Hence find the exact value of $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{9 \times 10}$. [E]

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QUESTION THREE

(a) Solve $\frac{4x+3}{x-5} - 2 = 0$. [A]

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(b) Solve $\frac{x+2}{3} - \frac{x-1}{4} = 2$. [A]

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(c) Make r the subject of $P = \frac{2r+5}{r-3}$. [M]

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(d) Solve $\frac{9}{x} - \frac{4}{x+1} = 2$.

[M]

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(e) A kayaker paddles 16 km up a river against a current of 2 km/h, then 16 km back down with the current. The whole trip takes 6 hours. Her speed in still water is v km/h.

(i) Show that $3v^2 - 16v - 12 = 0$.

[M]

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(ii) Find her speed in still water. Explain why there is only one answer.

[E]

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